Name: Henry Dean Group: with Chase Borjan TA/Lab Section: Harry

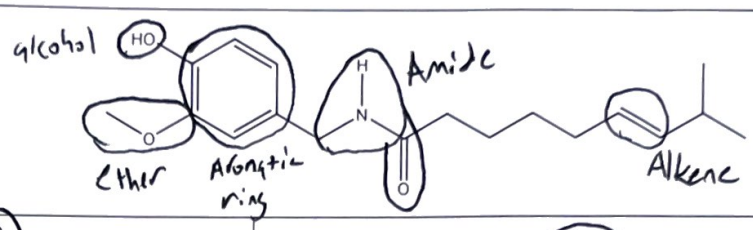
11 Organic Chemistry: Structures & IR Spectroscopy

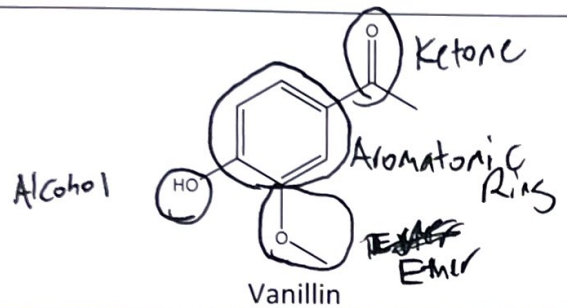
Part I: Review Functional Groups

1. Circle and identify all the following functional groups in the molecules below

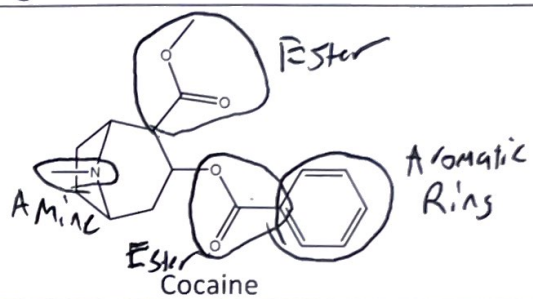
Alcohol, Aldehyde, Alkene, Alkyne, Amide, Amine, Aromatic ring, Carboxylic Acid, Ester, Ether, Ketone

Capsaicin



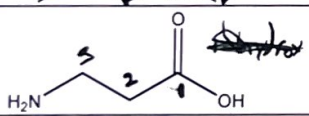


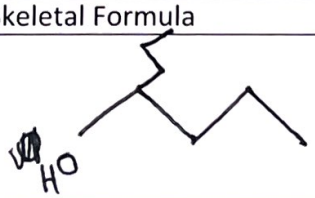
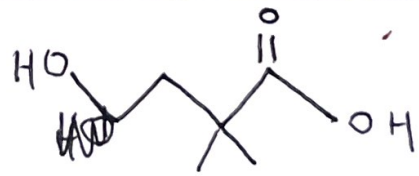
Vanillin



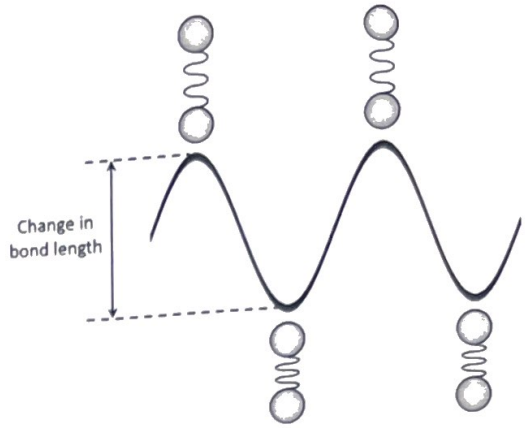
Cocaine

2. Practice naming and drawing the following molecules:

Molecular Structure	Name:
	~~4-hydroxy-2,4-pentanedione~~ 4-hydroxy-4-methyl-2-pentanone
	3-amino propanoic acid

Name	Skeletal Formula
2-propyl-1-pentanol	
5-hydroxy-2,2-dimethylpentanoic acid	

Introduction to Infrared (IR) Spectroscopy

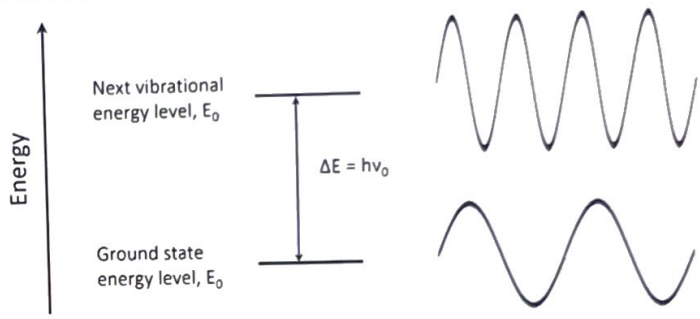


While representations of molecules make it seem like bond lengths and bond angles are fixed values, this is not the case. Bonds and bond angles within a molecule are dynamic; the atoms are constantly moving. A chemical bond is more like a spring than a rod; stretching and compressing in an oscillating pattern with an associated frequency. The vibrational energy of a chemical bond is dependent on the strength of the bond and the masses of the atoms involved.

$$E = \frac{h}{2\pi}\sqrt{\kappa\left(\frac{m_1 + m_2}{m_1 \times m_2}\right)}$$

Where h is Planck's constant, κ is the force constant (related to the strength of the bond), m_1 is the mass of the first atom and m_2 is the mass of the second atom in the bond.

Like electronic energy levels of molecular orbitals, vibrational energies of bond lengths and bond angles are quantized. If a photon with energy equal to $h\nu_0$ is absorbed, the vibrational energy will jump to the next energy level. Bond stretching and bending vibrational energies correspond to the Infrared region of the electromagnetic spectrum.



3. Calculate the value of $\left(\frac{m_1+m_2}{m_1 \times m_2}\right)$ for the following single bonds and place them in order of increasing vibrational stretching frequency:

$^{12}C-C^{12}$ 0.166 $^{12}C-O^{16}$ 0.1459 $^{12}C-H^{1}$ 1.09

C-O < C-C < C-H

4. The force constant of a bond is related to the strength of the bond. Place these bonds in order of vibrational stretching frequency based on their bond dissociation energies:

C ≡ C (837 kJ/mol) C = C (611 kJ/mol) C – C (347 kJ/mol) C-O (360 kJ/mol) C=O (736 kJ/mol)

Carbon-Carbon Bonds

C-C < C=C < C≡C

Carbon-Oxygen Bonds

C-O < C=O

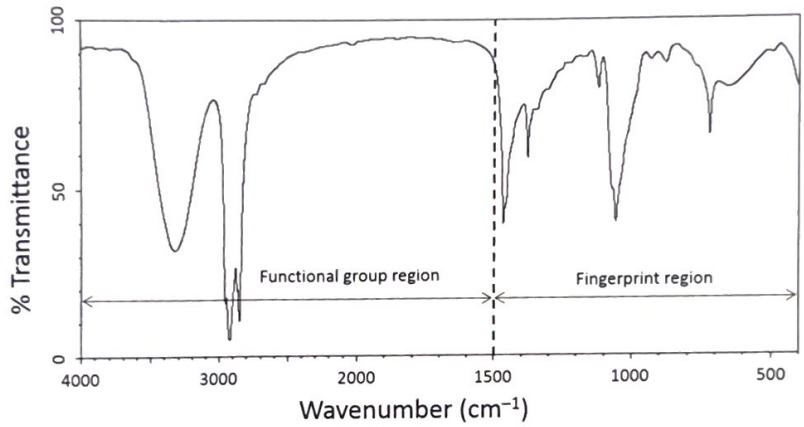
5. Do you expect bending a bond angle to require more or less energy than stretching / compressing a bond? Explain:

I expect stretching to require more energy because the chemical attraction must be directly opposed.

Bonds with polar character will absorb infrared light. If a bond has a dipole, as the bond length oscillates, the bond dipole will oscillate as well. The infrared light is actually interacting with the oscillating electric field of the bond, not the bond itself. Polar bonds within a molecule will absorb infrared light and non-polar bonds in polar molecules will also absorb infrared light. It is also possible that a non-polar molecule can absorb light due to temporary dipoles caused by asymmetric stretching or bending in the molecule. Bonds within a molecule that absorb Infrared light are known as infrared-active bonds.

Anatomy of an IR spectrum:

An infrared spectrum can be used to study the types of bonds within a molecule. A typical IR spectrum is shown below. The spectrum is a transmittance spectrum as opposed to an absorption spectrum like the ones you have worked with previously. The spectrum is plotted as % Transmittance vs wavenumber. Because it is a % transmittance spectrum, the "peaks" in the spectrum appear as troughs. For example, there are strong absorptions at ~2900 cm^{-1} and 3300 cm^{-1} in the spectrum below.



Wavenumber ($\tilde{\nu}$) is proportional to frequency and Energy and inversely proportional to wavelength and typically has units of cm^{-1}.

$$\tilde{\nu}\ (cm^{-1}) = \frac{\nu}{c} = \frac{E}{hc} = \frac{1}{\lambda}$$

Higher wavenumbers correspond to higher vibrational energies, therefore we expect to see the peaks of stronger vibrations on the left of the spectrum. Bond bending peaks typically appear to the right in the spectrum in the fingerprint region, less than 1500 cm^{-1}, while the characteristic peaks of functional groups typically appear between 1500-4000 cm^{-1}. When trying to identify functional groups in a molecule, focus first on the functional group region. The fingerprint region tends to be complicated with many overlapping weaker signals.

All IR spectra in this lab were obtained from the Spectral Database for Organic Compounds. SDBSWeb : https://sdbs.db.aist.go.jp (National Institute of Advanced Industrial Science and Technology, 4/1/2019)

Identifying functional Groups using IR

The bonds of functional groups result in characteristic peaks in an IR spectrum depending on the elements involved in the bond and the strength of the bond. In this activity, you are going to compare the IR spectra of molecules with different functional groups in order to identify the characteristic peaks of each functional group. Obtain a set of IR cards from your TA. Each IR card has the chemical structure one side and the corresponding IR spectrum and name on the other side. There are 16 cards per set.

Do not write on the IR cards as they will be used for later lab sections. Make all your notes in this handout. Start by sorting the cards by functional group.

6. Compare the IR spectra of the four alkanes. Identify two common peaks, one in the functional group region and one in the fingerprint region. These correspond to the C-H stretching and bending peaks. Which is which?

A peak at 3000 cm^{-1} and 1610 are common. The peak at 3000 is in the functional group region, the peak at 1610 is in the ~~functional~~ fingerprint. 3000 is stretching 1610 is bending.

7. Compare the structures of decane and decanol. What bonds are in decanol that are not in decane? What similarities do you observe in the spectra?

Decanol has an -OH group and decane doesn't. Decanol has C-O and O-H bonds. They share the first fingerprint peak and the first 3000 peak.

8. Compare the spectra of the three alcohols. What peak(s) are characteristic of the alcohol functional group? Which bonds do you attribute these peaks to? Refer back to question 3.

The peaks at 3400, ~~[illegible]~~ at 1450. The peak at 3400 is the O-H bond and the peak at 1450 is the C-O bond.

9. Compare carboxylic acids, ketones and aldehydes.

   a) What type of bond do all these functional groups contain? Identify the peak that corresponds to this bond.

All of these functional groups contain a C=O bond. This corresponds to the peak at 1700 cm^{-1}.

   b) Carboxylic acids and alcohols share a common type of bond. What is the bond? How is the shape of the peak of this type of bond different in the two functional groups?

They have an O-H bond. The peak of the O-H bond is shouldered and spikin.

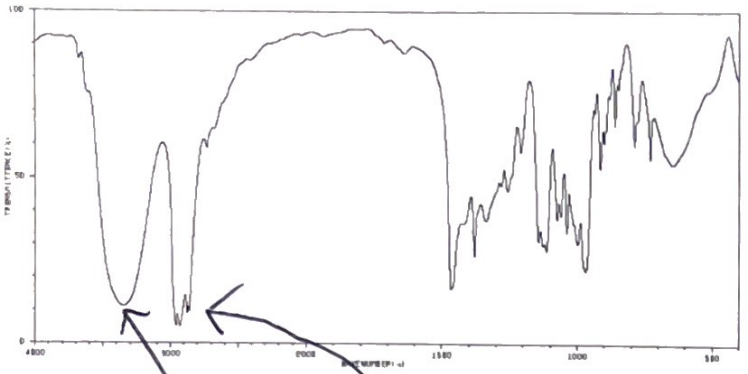
   c) How is the C-H stretching peak different in an aldehyde vs a ketone?

The aldehyde has a larger trough compared to the ketone at the C-H stretching peak.

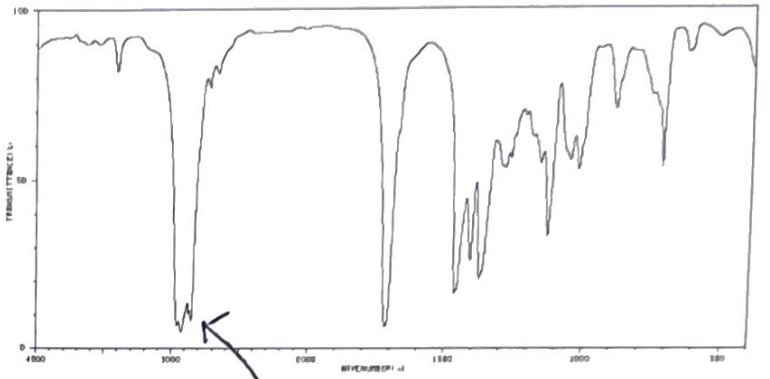
10. Compare the spectra of the ethers to the other oxygen containing functional groups. Do the ether spectra contain any of the characteristic peaks of the other functional groups?

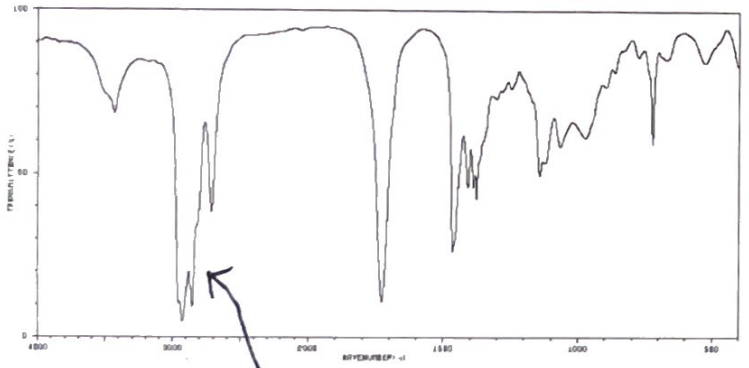
No, ether spectra don't have a C=O or O-H bond so they don't share any common peaks.

11. Based on your observations above, identify the functional groups in the molecules on the next page. Label the characteristic peaks that you used in the identification:

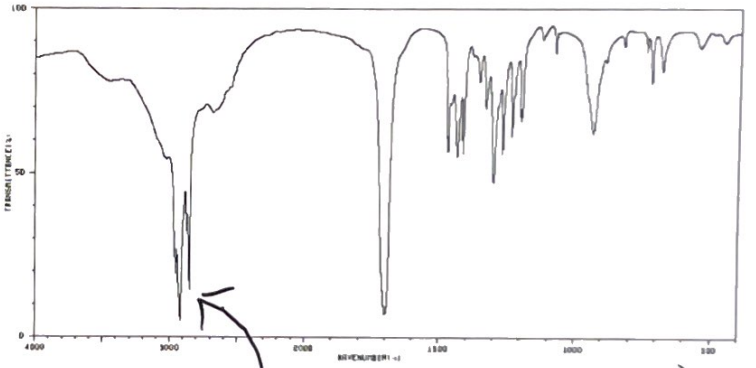


Functional Group: alcohol, Ketone	Functional Group: Ketone
Characteristic peak(s): 3300, 2400 cm^{-1}	Characteristic peak(s): 2400, 1750 cm^{-1}





Functional Group: Aldehyde,	Functional Group: ~~Aldehyde~~ Carboxylic Acid
Characteristic peak(s): ~~2800~~, 2400, 1750 cm^{-1}	Characteristic peak(s): 2400, 1750 cm^{-1}, 2750



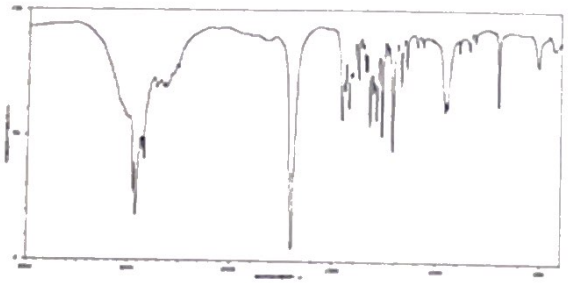
Which of the oxygen-containing functional group is not represented above? If you know a molecule contains an oxygen atom, how can you determine this functional group is present?

An ether is missing. We know its missing because an ether shows no O-H peak, no C-O peak, but a large peak at 1000-1200 cm^{-1}.

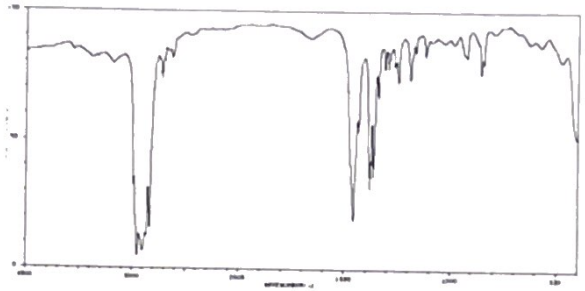
12. There are four models of molecules in the lab. For each molecule, A-D, use the model to answer the following questions:

- Identify the functional groups in the molecule
- Draw the structure of the molecule
- Name the molecule
- Identify which of the following IR spectra belongs to the molecule

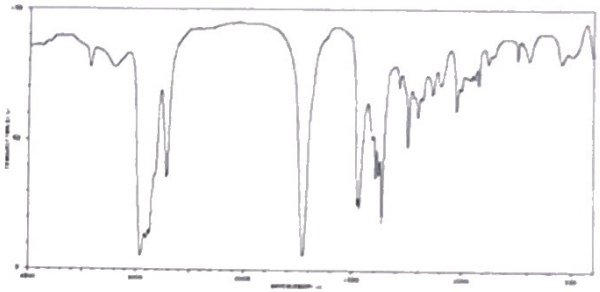
Spectrum 1



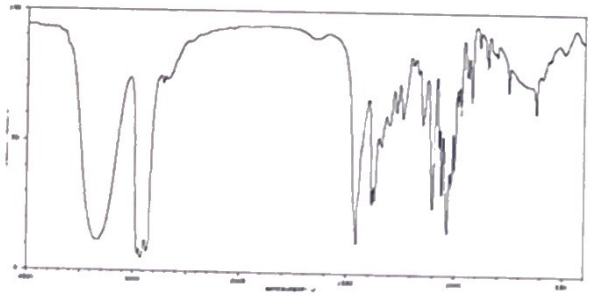
Spectrum 2



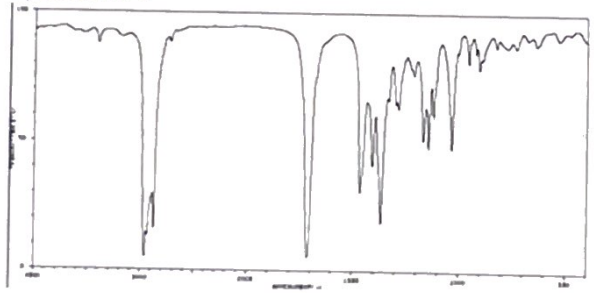
Spectrum 3

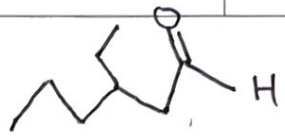
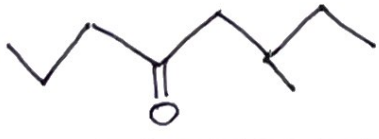


Spectrum 4



Spectrum 5



Molecule A	Structure	
	Name: 1-Propylcyclohexane	
	Functional Groups: N/A	IR Spectrum: 2
Molecule B	Structure 	
	Name: 3-ethylhexanal	
	Functional Groups: Aldehyde	IR Spectrum: 3
Molecule C	Structure	
	Name: 3-ethyl-5-methyl-1-cyclopentanol	
	Functional Groups: Alcohol	IR Spectrum: 4
Molecule D	Structure 	
	Name: 6-methyl-4-octanone	
	Functional Groups: Ketone	IR Spectrum: 5